Largest Perimeter Triangle 4

HackerRank

You are given an array nums of side lengths, return the largest perimeter of a triangle with a NON-ZERO area formed by three of these lengths.

However, you cannot just pick any three sides. The original array indices i, j, and k (where i < j < k) of the three chosen side lengths must form an Arithmetic Progression. This means the distance between the first and second chosen index must be exactly equal to the distance between the second and third chosen index (i.e., j − i = k − j).

Find the maximum perimeter of a valid triangle. If no valid triangle can be formed under these rules, return 0.

Input Format

The first line contains a integer, N, denoting the size of the array nums.

The second line contains N space-separated integers denoting the array, nums[i].

Constraints

1 <= N <= 10^5

1 <= nums[i] <= 10^8

Output Format

The output contains an integer value

Sample Input 0

```
5
10 5 100 5 10
```

Sample Output 0

```
0
```

Explanation 0

The valid triplets are: (0,1,2) -> 10, 5, 100 (Invalid) (1,2,3) -> 5, 100, 5 (Invalid) (2,3,4) -> 100, 5, 10 (Invalid) (0,2,4) -> 10, 100, 10 (Invalid) Since no valid triangle can be formed, the result is 0.

Sample Input 1

```
5
2 1 2 4 2
```

Sample Output 1

```
6
```

Explanation 1

The valid index triplets (i, j, k) that form an arithmetic progression are (0,1,2), (1,2,3), (2,3,4), and (0,2,4). Checking triplet (0,1,2): sides 2, 1, 2. Valid triangle, perimeter = 5. Checking triplet (1,2,3): sides 1, 2, 4. Invalid triangle (1+2 < 4). Checking triplet (2,3,4): sides 2, 4, 2. Invalid triangle (2+2 = 4). Checking triplet (0,2,4): sides 2, 2, 2. Valid triangle, perimeter = 6. The maximum valid perimeter is 6.